MANIPAL INSTITUTE OF TECHNOLOGY
BENGALURU
(A constituent unit of MAHE, Manipal)

Major Student Project Proposal

Design & Development of an Autonomous Stair-Climbing Robot for Surface Cleaning

Team Name : Stair Climber Vaccum Cleaner

Team Members : Krishna Saraf (Team Lead) ,Rishabh Singh, Suraj Kumar S, Bhoomika Babu, Shreya Basker

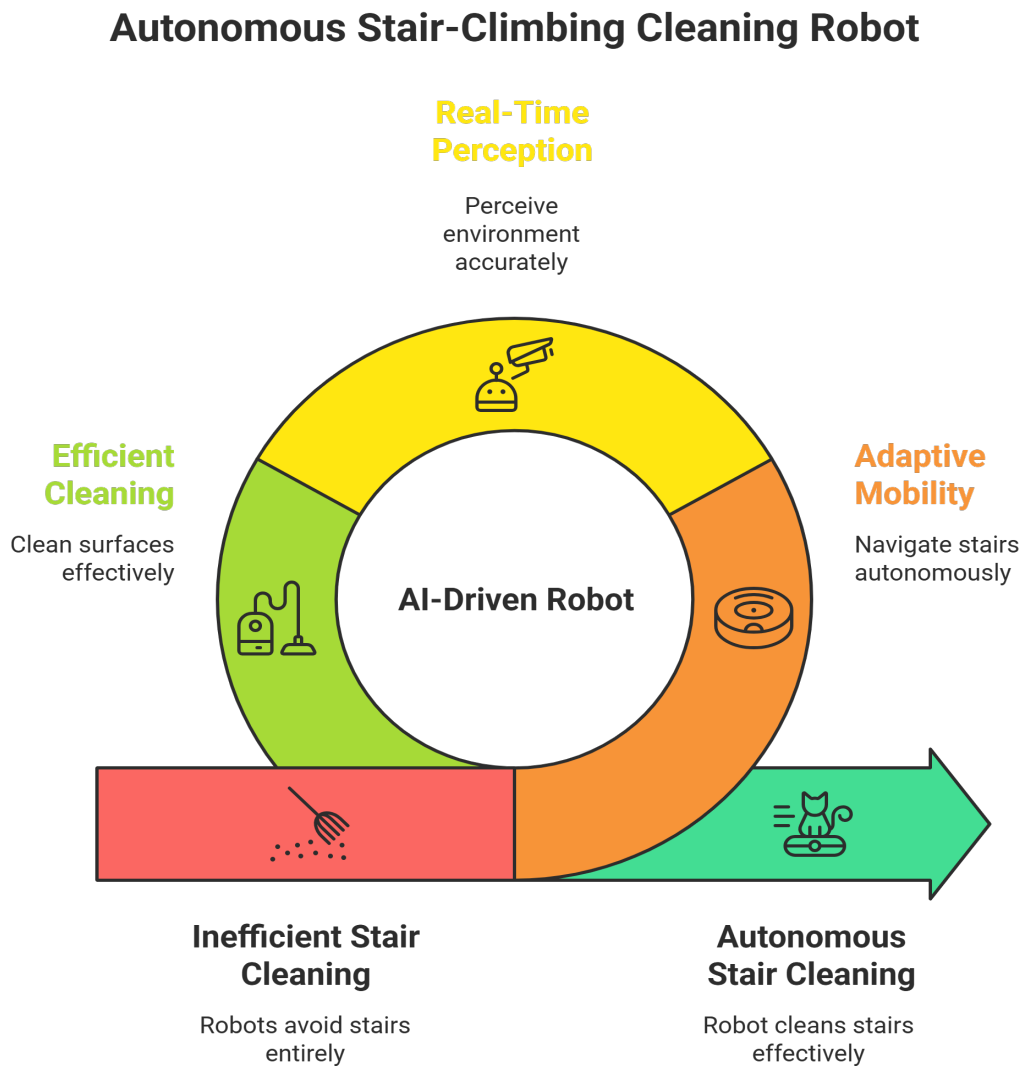
Subject Area : Robotics, Autonomous Mobility, Computer Vision, Sensor Fusion

Faculty Advisor : Dr. Megha. P. Arakeri

Background/Introduction

- Most robotic vacuum cleaners **avoid stairs**, leaving multi-level spaces uncleaned.
- Existing stair-climbing robots are **manual, bulky, and expensive**, with no cleaning integration.
- There is a **clear gap** for an affordable, compact, and autonomous stair-cleaning solution.

This project aims to **develop an AI-driven robot** capable of **autonomous stair climbing and surface cleaning** using **sensor fusion and adaptive mobility**.



Current players in the field





- **iRobot Roomba j7+** – Excels in smart floor cleaning but **cannot handle stairs**.
- **Roborock S7 MaxV Ultra** – Offers AI mapping and suction control yet **restricted to flat surfaces**.
- **Ecovacs Deebot U2 Pro** – **Budget-friendly** but lacks adaptability on uneven terrains.
- **Honda E2-DR** – Demonstrates **stair-climbing robotics**, but **no cleaning functionality** and extremely high cost.
- **ARGO J5 & Scalamobil S35** – Focus on **stair mobility assistance**, not autonomous cleaning; **manual and industrial** in nature





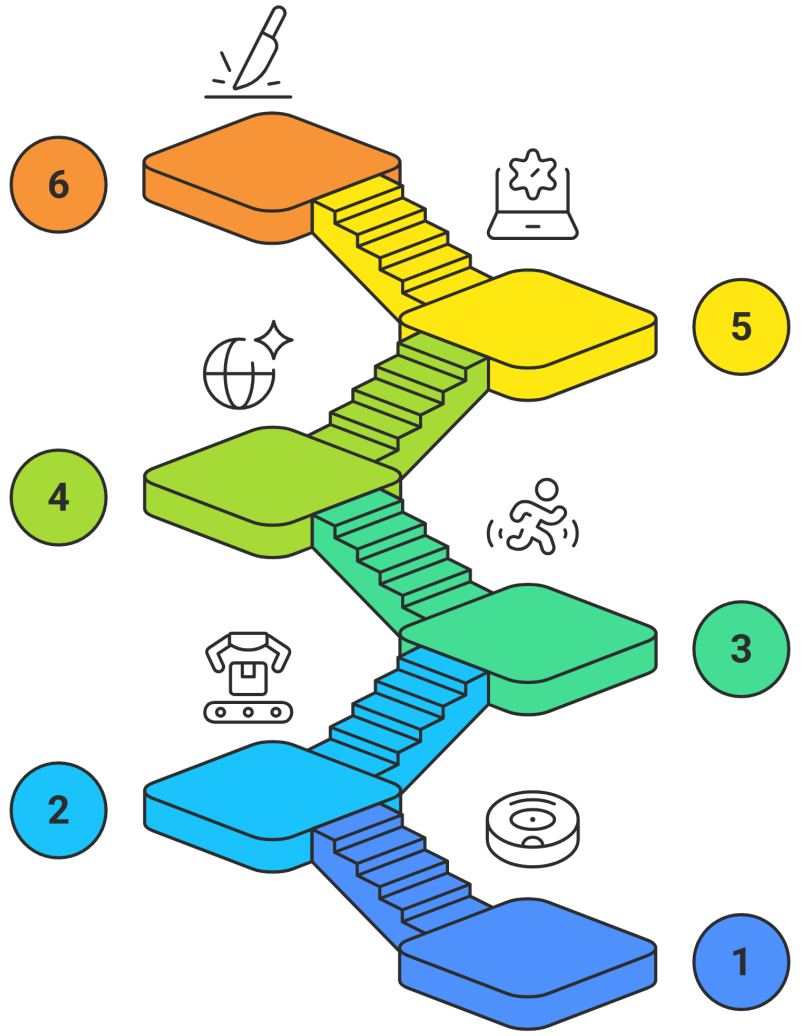
🔷 **Key Takeaway:**
Current solutions either **clean or climb — never both.**
Our project bridges this gap with an **AI-driven, autonomous stair-climbing cleaning robot** that combines mobility, perception, and efficiency in one system.

Project Objectives

🎯 **Vision:**

To pioneer a **cost-effective, AI-powered robotic solution** that bridges the gap between household cleaning and autonomous stair mobility.

Achieving Autonomous Stair-Climbing Robot



Develop Robot (1)
Create an intelligent, autonomous robot capable of cleaning and climbing stairs.

Design Chassis (2)
Design a modular, adjustable chassis for balance and stability.

Implement Sensors (3)
Integrate LiDAR, camera, and ultrasonic sensors for environmental awareness.

Integrate AI (4)
Implement AI algorithms for stair recognition and navigation.

Optimize Power (5)
Optimize power and performance through efficient hardware design.

Validate Prototype (6)
Validate the prototype through rigorous testing on diverse surfaces.

Identified Competitions

- Mainly aiming for Patenting.
- Competitions:

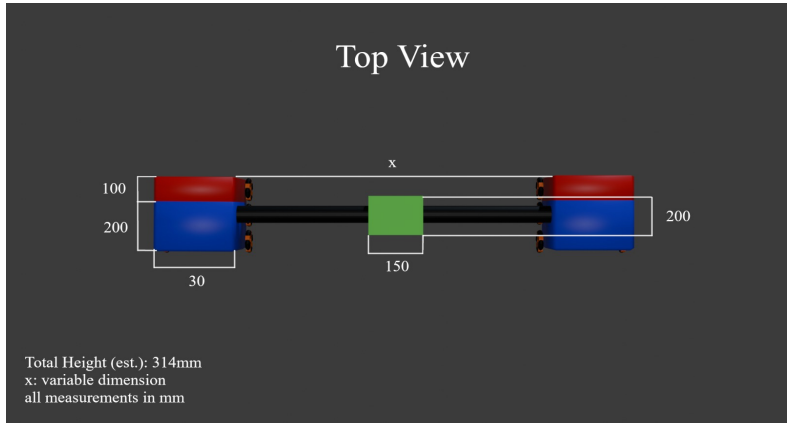
1. JMT (Junior Make-A-Thon) - Shaastra, IIT Madras
2. IIM Bangalore's Social Innovation Ideathon
3. Technoxian World Robotics Championship Series
4. FIRST Tech Challenge (FTC) India
5. Robofest India
6. Robotix (IIT Kharagpur / Kshitij festival)

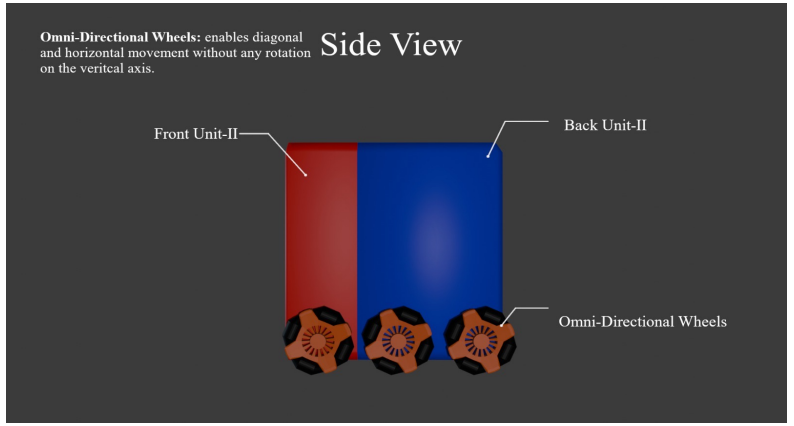
Methodology

Objective 1:

To design a modular, autonomous stair-climbing cleaning robot that integrates mechanical innovation, an adjustable chassis, and a robust cleaning mechanism for efficient debris removal and adaptive stair navigation

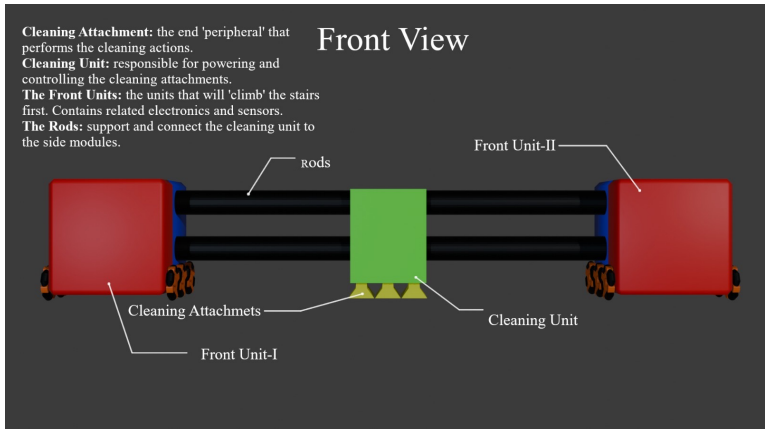
- The robot consists of two parallel chambers placed side by side across the stair width, connected by an adjustable central frame that adapts automatically to different stair widths for balance and grip.
- Each chamber has two sections – front (red) and back (blue) – which work alternately during climbing. The front section lifts first, followed by the back, ensuring smooth, step-by-step motion.
- The adjustable frame, powered by linear actuators or telescopic rods, regulates the spacing and stabilizes both chambers during motion.
- Omni-directional wheels enable movement in any direction, simplifying turning and platform navigation.
- Rotating brushes are fitted below the back sections for underbody cleaning, while the central unit between the chambers performs vacuuming and mopping simultaneously.
- The modular, low-weight aluminum and 3D-printed structure ensures stability, adaptability, and easy maintenance.



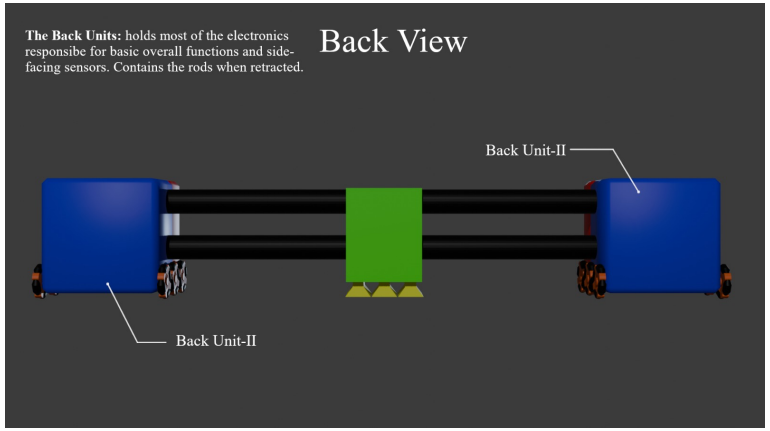


Objective 2:

To implement an effective perception system using camera and LiDAR sensors for real-time detection of stair dimensions, surface types, and obstacles, enabling precise and adaptive movement

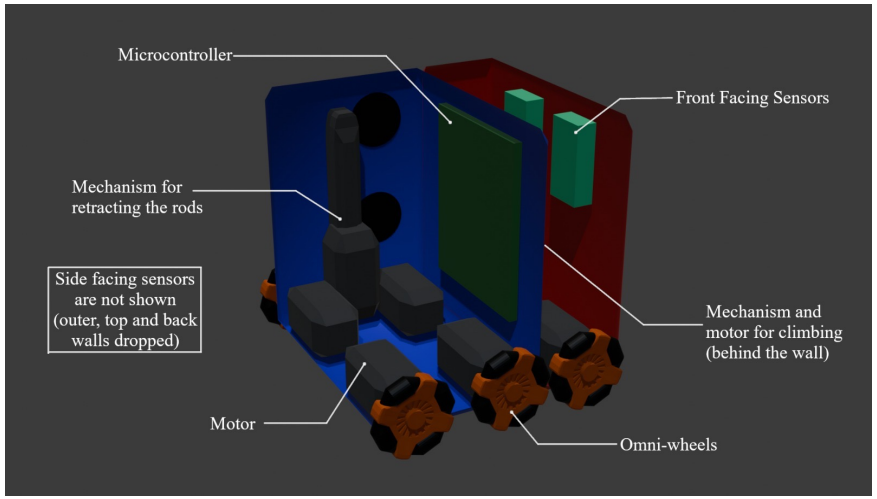


- The robot uses a multi-sensor fusion system combining LiDAR, AI camera, ultrasonic, infrared, and IMU sensors.
- LiDAR measures stair height and depth for precise step recognition, while the camera (with CNN model) identifies surface type (tile, wood, carpet, etc.).
- Ultrasonic and IR sensors detect nearby obstacles, edges, and walls, preventing collisions during climbing.
- The IMU tracks tilt and orientation to maintain balance on uneven stairs.
- Sensor data is processed through Kalman filtering to generate accurate 3D environmental maps.
- Based on detected geometry, the control unit dynamically adjusts chamber spacing, motor torque, and suction power for optimal performance.



Objective 3:

To develop AI algorithms for stair-type classification and optimal climbing strategy prediction, supporting intelligent navigation and real-time decision-making through microcontroller-based control and efficient power management



- Raspberry Pi handles AI processing (vision, SLAM, and strategy prediction), while Arduino Mega manages motor control and sensor input.
- The AI uses:

  CNN for stair-type and surface classification.

  Reinforcement Learning (RL) for predicting the safest and most efficient climbing strategy.
- PID control ensures smooth, stable motion, while inverse kinematics synchronize omni-wheel movements for precise direction control.
- A 14.8V Li-ion battery powers the system, with dual regulators isolating motor and logic circuits for safety.
- The power module intelligently adjusts suction, mop speed, and actuator torque based on workload and dirt density, improving efficiency and battery life.

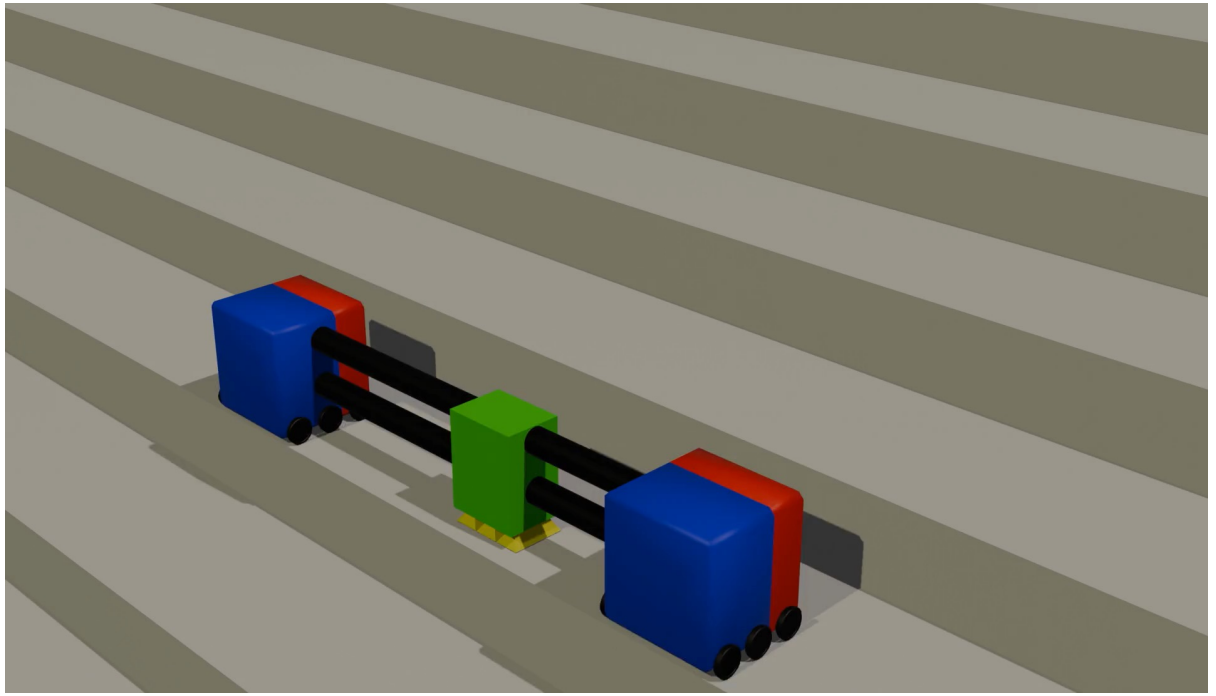
Objective 4:

To test and validate the robot's performance in various indoor environments, assessing cleaning efficiency, climbing reliability, obstacle handling, and energy consumption under real-world conditions

- The prototype is tested on varied staircases (wood, tile, marble, and irregular steps) to validate real-world performance.

- Key metrics include:

1.Climbing success rate (%)
2.Cleaning efficiency (%)
3.Stability and tilt deviation (°)
4.Obstacle detection accuracy (%)
5.Power consumption (Wh/cycle)



- Performance Tests:

1.10-step stair climb trials to check balance and coordination.
2.Dust removal and mopping tests to measure cleaning coverage.
3.Obstacle placement experiments for vision and LiDAR response.
4.Battery endurance tests under continuous operation.

- Results are analyzed to fine-tune motor torque, climbing speed, suction strength, and AI navigation for maximum reliability and energy efficiency.

Timeline –Semester wise

Phase & Task	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Phase 1: Research & Planning																
Phase 2: Design & Simulation																
Phase 3: Electronics Development																
Phase 4: Prototyping & Assembly																
Phase 5: Testing & Optimization																
Phase 6: Finalization & Documentation																

Expected Outcomes/ Deliverables

- **Fully functional stair-climbing vacuum cleaner prototype capable of autonomous navigation and cleaning on stairs.**
- **Detailed CAD designs and mechanical drawings of the main and side units.**
- **Complete PCB schematics and embedded control code for hardware integration.**
- **Performance evaluation reports, including testing data, optimization results, and analysis of cleaning efficiency and stability.**
- **Documentation of AI algorithms for stair detection, path planning, and sensor fusion.**

Project Relevance and Applications

- **Staircases remain a critical gap in home automation, as 98% of robotic vacuums (e.g., Roomba j7+, Roborock S7 MaxV Ultra) avoid stairs entirely.**
- **The project addresses this by creating a dual-unit robotic system that integrates stair climbing with vacuuming/mopping, a feature absent in commercial solutions like the Migo Ascender (still unreleased as of 2025).**
- **With 63% of Indian households having multi-level homes, this technology bridges the divide between floors, eliminating manual labour for stair cleaning.**

Applications of the Project:

- **Accessibility: Enhances mobility for elderly/disabled individuals by automating stair cleaning, reducing fall risks during manual efforts**
- **Cost efficiency: Targets affordability (prototype budget: 35,000) versus commercial stair climbers like ARGO J5(~24.9 lakh)**
- **Sustainability: Reduces reliance on multiple devices (eg. Separate vacuums per floor), cutting e-waste**
- **Industrial logistics: Adaptable for transporting goods in warehouses with staircases**

Budget proposal

1st Six Months/ First Phase			
S.No	**List of Non-recurring expenditure**	**Total Qty**	**Total Amount**
1	Arduino Mega	4	4000
2	AI machine vision sensor	2	14000
3	Mecanum Wheels	24	32900
4	Motor drivers (BTS7960)	12	3100
5	Torque Motors	6	8000
6	Drive Motors	12	20000
7	Battery	4	40000
8	Wires(all types)	400	2000
9	Shaft couple	12	800
10	Gears,rack and pinion	8	12000
11	Lidar sensor module	2	8000
12	Touch Sensor	8	1600
13	Ultrasonic Sensor	4	1800
14	Chassiss and Mechanical Parts: 3d printing and Aluminium		80000
	Total Non-recurring amount		2,28,200
S.No	**List of Recurring expenditure**	**Total Qty**	**Total Amount**
1	*Chemicals/ Glassware*		3500
2	*Glassware*	-	-
3	*Electrical & Electronics Spare Parts*		12000
4	*Mechanical Spare Parts*		10000
	Total Recurring amount		25,000
	1st Phase Total Non-recurring and Recurring		2,53,700

2nd Six Months/ Second Phase			
S.No	**List of Non-recurring expenditure**	**Total Qty**	**Total Amount**
1	Raspberry pi 5 model 16gb ram	4	60000
2	Raspberry pi camera module	2	6124
3	NVIDIA Jetson Nano B01 4GB Developer Kit	2	40000
4	Intel realsense d4351	2	80000
	Total Non-recurring amount		1,86,124
S.No	**List of Recurring expenditure**	**Total Qty**	**Total Amount**
1	*Chemicals/ Glassware*		-
2	*Glassware*		-
3	*Electrical & Electronics Spare Parts*		6176
4	*Mechanical Spare Parts*		8000
	Total Recurring amount		14,176
	2nd Phase Total Non-recurring and Recurring		2,00,300

3rd Six Months/ Third Phase			
S.No	**List of Non-recurring expenditure**	**Total Qty**	**Total Amount**
1	Vacuum motor	4	21200
2	Post and pre motor hepa filters	4	5600
3	Dust bags	4	2000
4	Hose pipe , mop	4	4000
	Total Non-recurring amount		32,800
S.No	**List of Recurring expenditure**	**Total Qty**	**Total Amount**
1	*Chemicals/ Glassware*		2000
2	*Glassware*		-
3	*Electrical & Electronics Spare Parts*		4000
4	*Mechanical Spare Parts*		3600
	Total Recurring amount		9600
	3rd Phase Total Non-recurring and Recurring		42,400

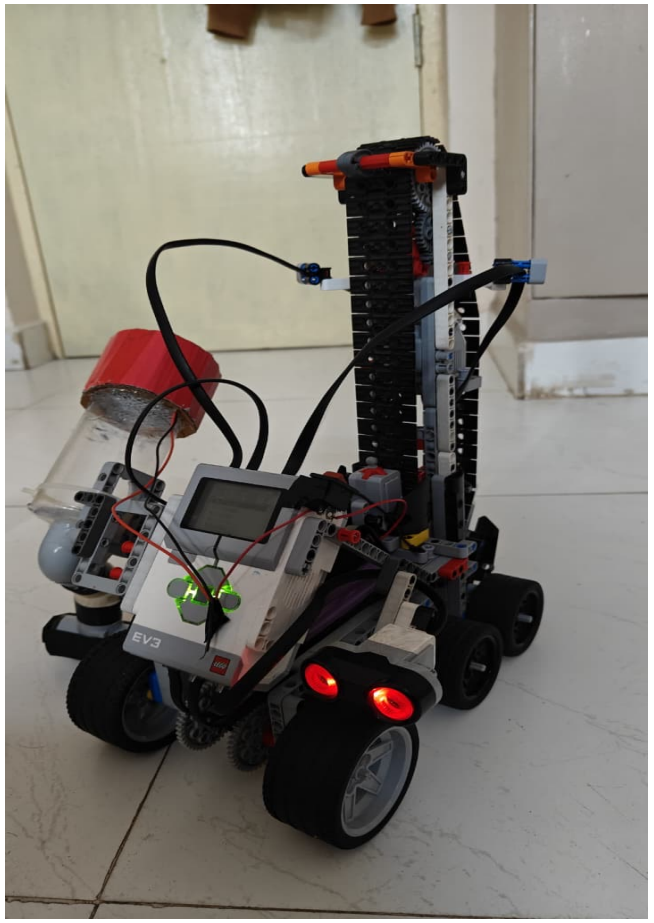
A)	**Total Amount (1st Six months/First Phase)**	2,53,700
B)	**Total Amount (2nd Six months/Second Phase)**	2,00,300
C)	**Total Amount (3rd Six months/Third Phase)**	42,400
	GRAND TOTAL	4,96,400

Budget for AY 25-26

1st Six Months/ First Phase			
S.No	**List of Non-recurring expenditure**	**Total Qty**	**Total Amount**
1	Arduino Mega	4	4000
2	AI machine vision sensor	2	14000
3	Mecanum Wheels	24	32900
4	Motor drivers (BTS7960)	12	3100
5	Torque Motors	6	8000
6	Drive Motors	12	20000
7	Battery	4	40000
8	Wires(all types)	400	2000
9	Shaft couple	12	800
10	Gears,rack and pinion	8	12000
11	Lidar sensor module	2	8000
12	Touch Sensor	8	1600
13	Ultrasonic Sensor	4	1800
14	Chassiss and Mechanical Parts: 3d printing and Aluminium		80000
	Total Non-recurring amount		2,28,200
S.No	**List of Recurring expenditure**	**Total Qty**	**Total Amount**
1	*Chemicals/ Glassware*		3500
2	*Glassware*	-	-
3	*Electrical & Electronics Spare Parts*		12000
4	*Mechanical Spare Parts*		10000
	Total Recurring amount		25,000
	1st Phase Total Non-recurring and Recurring		2,53,700

Any Other Information/Work done till date/Achievements of team

Video_link



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